



Thicken+

Onshape's Thicken, with the two thicknesses named 'toward reference' and 'away from reference' instead of by the face normal -- the boolean step unchanged.



What it does

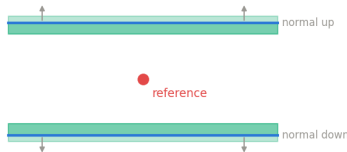
The built-in Thicken's Thickness 1 goes along the face normal, so the same feature thickens outward on one model and inward on another. Thicken+ decides per surface body which way is 'toward the reference' and puts the Toward thickness there. Everything else -- New / Add / Remove / Intersect, merge scope -- is the built-in.

Use it when

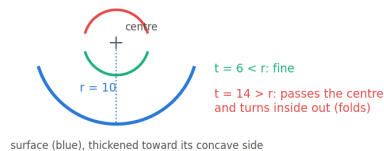
- Surfaces of mixed orientation must thicken the same way
- A surface's normal may flip upstream
- You want a clear error where a thicken would fold, with its location

Toward / away decided per surface body

Toward = 5 (green), Away = 3 (light); both sheets thicken toward the reference, whatever their normals



Curvature check: concave radius < thickness -> error with its location



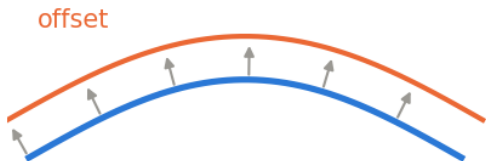
Left: both sheets thicken toward the reference whatever their normals. Right: the curvature check.



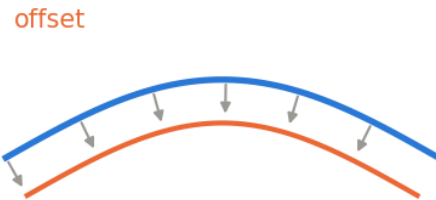
Why a reference instead of a flip

A flip names a side relative to a normal; a reference names it in the model

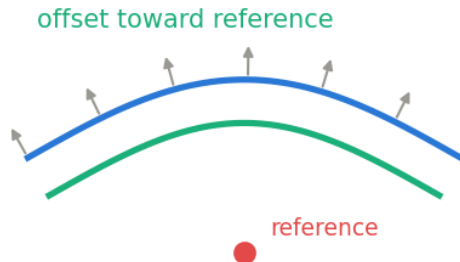
Built-in, normal up:
'offset along normal' goes up



Same model after an upstream change
flips the normal: now it goes DOWN



Offset+ 'toward reference':
goes toward the point, either way



A flip is about a normal

'Front', 'opposite direction', 'Thickness 1'
all mean: relative to the normal -- and
normals change freely upstream.

A reference is about the model

Pick geometry on the side you mean. The
choice survives any change of
orientation.

Drop-in

With no reference, the feature behaves
exactly like the built-in.



The dialog

T2 Thicken+ two sheets ... ✓ ✗

1 **New** Add Remove Intersect

2 Faces and surfaces to thicken

Surface 17 ✕

Surface 18 ✕

3 Side reference 

T2 reference point (9000,... ✕

4 Toward reference 2 mm

5 Away from reference 3 mm 

7 ☐ Keep tools

8 > Advanced

 Final

#	Parameter	What it means
1	Creation type	New / Add / Remove / Intersect -- the built-in boolean step, unchanged.
2	Faces and surfaces to thicken	Surfaces or faces; each owner body is thickened on its own side.
3	Side reference	Geometry on the side to thicken toward -- best a mate connector or vertex near the surface. Empty = along the normals.
4	Toward reference	Thickness on the reference's side.
5	Away from reference	Thickness on the other side.
6	Swap sides	The arrow: exchange the two thicknesses.
7	Keep tools	Keep the input surfaces.
8	Advanced	Check curvature (on): report where a concave radius is smaller than the thickness. Print details.



Example: opposite normals, same result

**T2: two sheets at z +50 with OPPOSITE normals, reference at the centre;
Toward 2 mm, Away 3 mm (section)**


top: 2 toward the centre, 3 away
-> z 48..53

reference (z 0)

bottom: the same, mirrored
-> z -53..-48


Test T2: sheets at z +50 and -50 with opposite normals, reference at the centre, Toward 2, Away 3

Both sheets put 2 mm toward the centre and 3 mm away: top z 48..53, bottom z -53..-48 -- mirror images, although their normals point opposite ways. T3 (Swap sides) gives 47..52 / -52..-47. T6: Add into a cube, 5 mm up -> one part, volume 1 018 000 mm³.



The curvature check

Toward / away decided per surface body

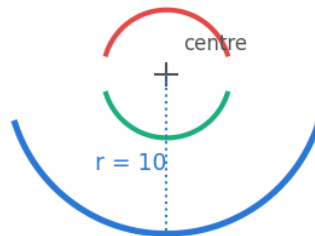
Toward = 5 (green), Away = 3 (light): both sheets thicken toward the reference, whatever their normals



reference



Curvature check: concave radius < thickness -> error with its location



$t = 6 < r$: fine

$t = 14 > r$: passes the centre and turns inside out (folds)

surface (blue), thickened toward its concave side

Why thickens fail

Thickening toward a concave side by more than its radius turns the offset inside out.

Found before the kernel fails

Each face is sampled on a 9 x 9 grid; the first place that would fold is marked red, with its radius.

Tests T4 / T5

Half cylinder $r = 10$ thickened toward its axis: 15 mm -> the error with location; 5 mm -> a clean shell.



Outputs and messages

Published for Extract variables

Key	What it holds
output	The thickened bodies (or the parts they merged into).
towardFaces	The faces on the reference's side (at the Toward thickness).
awayFaces	The faces on the other side (on the input surface when Away is 0).
sideFaces	The faces around the edges of the thickened surfaces.

Messages

	Message / what to do
Error	Both thicknesses are zero.
Error	The side reference lies on surface k Move the reference off the surface.
Error	Surface k is concave with radius r mm at (x, y, z) ... the thicken would fold there. Reduce that side's thickness below r, or swap sides. The red point marks the place.



Tips and limits

✓ **Use it instead of Thicken downstream**
Surfaces from Split+, Mutual Trim+ or Offset+ keep their inputs' orientation; Thicken+ does not care which way they face.

✓ **Faces stay named through the boolean**
towardFaces / awayFaces / sideFaces still resolve after Add merges the result into a part.

! **Orientation does not apply**
Its outputs are solids, which always face outward -- Orient to reference is for surfaces.



Where it lives

Document	Reference_Side_Features (22764764a00a7f607dbc1c4d)
Tab	thicken_plus (uses reference_side_utils)
Tests	Reference side tests studio: T1-T6 (T4 must error) - devtools/onshape/check_reference_side_tests.py (34/34)
Examples	topsheet_surf studio (the user's derive of Design_Master topsheet and sidewall bodies)
Explainer	docs/explainers/reference_side/reference_side_explained.md, section 2.4
Related	Split+, Mutual Trim+, Offset+, Thicken+, Orient to reference (same reference rule); Extract variables